



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

AI Agents for Business

TGS Ref No: TGS-2023018987

Conducted by

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Version v1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
Legacy	Prior to this release	Existing TMS papers: Fine-Tuning LLM Models and RAG; version/date not stated in the recovered papers.	Existing course record
1.0	7 September 2026	Rebuilt as AI Agents for Business: 220 visual slides, detailed LG, LP, 12 self-contained labs and aligned WA/PP. Original K1-K14/A1-A6 and 60/90 minute timings preserved.	Tertiary Infotech Academy Pte Ltd

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Course identity and delivery allocation

AI Agents for Business | TGS-2023018987 | v1.0. Two days, 16 contact hours: 810 instructional minutes plus 150 assessment minutes. Breaks excluded from contact hours. The original WA/PP papers specify 60/90 minutes; these are retained. The website's 120-minute assessment description requires administrative reconciliation; no TMS duration field is changed by this release.

Day	Instruction	Assessment	Total contact
1	480 minutes	0	480 minutes
2	330 minutes	150 minutes	480 minutes
Total	810 minutes	150 minutes	960 minutes

Day 1 - 1 to 110

Session	Minutes	Slides	Method / evidence
Topic 1 teaching + administration	120	12-60	Mechanism explanation, worked artifacts, comparison and discussion
Lab 1: Workflow business case	30	52-54	Guided practice, evidence capture and peer review
Lab 2: Tokenize business tickets	45	55-57	Guided practice, evidence capture and peer review
Lab 3: Embeddings and similarity	40	58-60	Guided practice, evidence capture and peer review
Topic 2 teaching + administration	120	61-109	Mechanism explanation, worked artifacts, comparison and discussion
Lab 4: Grounded policy retrieval	35	101-103	Guided practice, evidence capture and peer review
Lab 5: Bounded tool agent	45	104-106	Guided practice, evidence capture and peer review
Lab 6: Support approval workflow	45	107-109	Guided practice, evidence capture and peer review

Insert meal and comfort breaks outside the 480 contact minutes. The trainer allocates the teaching/administration block across the topic slides and required opening/closing pages. Slide counts are 110 per day; depth is prioritised through worked examples rather than reading every slide aloud.

Day 2 - 111 to 220

Session	Minutes	Slides	Method / evidence
Topic 3 teaching + administration	75	119-167	Mechanism explanation, worked artifacts, comparison and discussion
Lab 7: Sales evidence and consent	25	159-161	Guided practice, evidence capture and peer review
Lab 8: Invoice exception agent	30	162-164	Guided practice, evidence capture and peer review
Lab 9: HR and multi-agent handoffs	25	165-167	Guided practice, evidence capture and peer review
Topic 4 teaching + administration	75	168-216	Mechanism explanation, worked artifacts, comparison and discussion
Lab 10: Train a support classifier	50	208-210	Guided practice, evidence capture and peer review
Lab 11: Agent evaluation scorecard	25	211-213	Guided practice, evidence capture and peer review
Lab 12: Governance and incident recovery	25	214-216	Guided practice, evidence capture and peer review
Written Assessment	60	217-219	Individual open-book knowledge paper, K1-K14
Practical Performance	90	217-219	Individual code and evidence, A1-A6

Insert meal and comfort breaks outside the 480 contact minutes. The trainer allocates the teaching/administration block across the topic slides and required opening/closing pages. Slide counts are 110 per day; depth is prioritised through worked examples rather than reading every slide aloud.

Lab facilitation and acceptance

Lab 1: Workflow business case - slides 52-54

Select a bounded business workflow and calculate net capacity released.

Alignment: K1, B02, B09, B10

Evidence: Support baseline releases 83.33 hours before additional overhead. The proposed pilot includes an owner and an explicit action boundary. A sensitivity result and quality floor accompany the financial claim.

Resources: labs/lab-01-workflow-business-case; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 2: Tokenize business tickets - slides 55-57

Use a real Hugging Face tokenizer and explain IDs, masks, subwords and unknown terms.

Alignment: K3, K6, K7; A1

Evidence: Real Hugging Face BertTokenizerFast output exists. Token IDs and masks have equal length for every example. The learner explains unknown tokens, subword boundaries and truncation.

Resources: labs/lab-02-tokenize-business-tickets; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 3: Embeddings and similarity - slides 58-60

Represent tokens with a transformer and compare cosine with Euclidean similarity.

Alignment: K2, K4, K8, K10, K11; A2, A3

Evidence: The real encoder output shape is recorded. $\text{Cosine}(q,d1)=1$ and $\text{cosine}(q,d2)$ is about 0.7071. The report distinguishes encoder representation from autoregressive text generation.

Resources: labs/lab-03-embeddings-and-similarity; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 4: Grounded policy retrieval - slides 101-103

Retrieve supporting policy clauses and abstain when evidence is absent.

Alignment: K8, K12, K14, B03-B05

Evidence: Every asserted rule has a supporting policy ID. The missing shipping guarantee triggers abstention. Injected document instructions do not grant authority.

Resources: labs/lab-04-grounded-policy-retrieval; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 5: Bounded tool agent - slides 104-106

Run a model-directed tool loop with schema checks and a strict read-only boundary.

Alignment: B01, B02, B06-B08, B12

Evidence: Allowed lookup returns O-104 with a source record. Cross-customer lookup and write tools are rejected. The final answer relies on actual tool results and stops within six rounds.

Resources: labs/lab-05-bounded-tool-agent; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 6: Support approval workflow - slides 107-109

Prepare a refund response and demonstrate approval-bound state transitions.

Alignment: K1, K12, B08, B13-B15

Evidence: Missing receipt does not become an approved refund. The repeated request does not create a second ledger operation. Approval is tied to the exact action payload.

Resources: labs/lab-06-support-approval-workflow; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 7: Sales evidence and consent - slides 159-161

Produce sourced lead briefs and consent-respecting outreach drafts.

Alignment: K1, B16-B18

Evidence: L-01 score is 75; L-02 score is 40. No-contact records are marked `research_only`. Every marketing claim is from the approved claim file.

Resources: labs/lab-07-sales-evidence-and-consent; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 8: Invoice exception agent - slides 162-164

Perform deterministic three-way matching and explain exceptions with evidence.

Alignment: K1, B19, B20

Evidence: INV-100 variance is 50.00. Duplicate and missing-PO records are flagged. The memo does not authorise payment.

Resources: labs/lab-08-invoice-exception-agent; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 9: HR and multi-agent handoffs - slides 165-167

Coordinate two bounded specialist roles and reconcile their outputs.

Alignment: K12, B21-B24

Evidence: Operations staff do not receive manager-only approvals. The handoff preserves `task_id` and source references. The final checklist resolves or escalates conflicting evidence.

Resources: labs/lab-09-hr-and-multi-agent-handoffs; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 10: Train a support classifier - slides 208-210

Train and compare real Hugging Face classifiers and a transparent baseline.

Alignment: K5, K9, K13; A4, A5, A6

Evidence: Training runs use real gradients and save a reloadable model. Training/validation/test IDs and texts are disjoint. Two strategies and a count baseline are compared; selection uses validation only.

Resources: labs/lab-10-train-a-support-classifier; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 11: Agent evaluation scorecard - slides 211-213

Compute task-quality and cost metrics with explicit denominators.

Alignment: K9, K13, B25-B30

Evidence: Precision=0.75, recall=0.90, accuracy=0.92. Cost per success includes the cost of failed runs. Release recommendation includes failure evidence, not only averages.

Resources: labs/lab-11-agent-evaluation-scorecard; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Lab 12: Governance and incident recovery - slides 214-216

Test denied actions, approval binding and a stop/recovery plan.

Alignment: B31-B38

Evidence: Unapproved writes and cross-customer reads are denied by code. Stop state prevents execution regardless of model output. Recovery checks transaction state before retry and assigns an owner.

Resources: labs/lab-12-governance-and-incident-recovery; README/PROMPTS PDFs, synthetic data, scripts and output template. Trainer observes actual execution and independently checks the learner's interpretation.

Assessment mapping

Criterion	WA question / PP task	Taught evidence
K1	WA Question 1	Slides 13-15
K2	WA Question 2	Slides 16-18
K3	WA Question 3	Slides 19-21
K4	WA Question 4	Slides 22-24
K5	WA Question 5	Slides 25-27
K6	WA Question 6	Slides 28-30
K7	WA Question 7	Slides 31-33
K8	WA Question 8	Slides 34-36
K9	WA Question 9	Slides 37-39
K10	WA Question 10	Slides 40-42

Criterion	WA question / PP task	Taught evidence
K11	WA Question 11	Slides 43-45
K12	WA Question 12	Slides 46-48
K13	WA Question 13	Slides 49-51
K14	WA Question 14	Slides 62-64
A1	PP Task 1	Lab 2
A2, A3	PP Task 2	Lab 3
A4	PP Task 3	Lab 10
A5, A6	PP Task 4	Lab 10

Preparation and release controls

Verify package version, install the three model-lab environments and execute all verification scripts before delivery. Keep answer keys private. Use the current LMS candidate papers and preserve the 60/90-minute timings. Record technical failures and provide an equivalent controlled rerun rather than fabricating evidence.